Worksheet 6

Exercise 1: Webcams

In a research and art project titled “Watching the World” (https://webcamaze.engineering.zhaw.ch/), AI is used to extract interesting images from publicly accessible webcams. In a study, participants were asked about various aspects of randomly selected webcam images. The goal is to determine what makes an image interesting. The study data is available in the file `interesting.rda`.

a) Estimate the causal structure using the PC algorithm. Visualize the estimated CPDAG. Use alpha = 0.0001.

b) Test the sensitivity of the PC algorithm to different α values.

c) Have a look at the estimated subgraph for the 5 variables: `storytelling`, `shareabilty`, `hold_attention`, `memorability` and `interestingness`. Report all DAGs that belong to the Markov equivalence class of the estimated CPDAG subgraph.

d) Estimate the causal structure using the LiNGAM algorithm under the assumption of non-Gaussian errors. Visualize the estimated DAG.

e) Compare the two estimated graphs from (a) and (b).

f) Based on the results of the LiNGAM algorithm, estimate the following total causal effects:

- of `novelty` on `surprise`
- of `storytelling` on `interestingness`
- of `memorability` on `interestingness`
- of `suprise` on `interestingness`

Exercise 2: Travel Time

The dataset `bus.rda` contains simulated travel times for a long-distance bus journey from Zurich to Rome.

a) Estimate the causal structure using the LiNGAM algorithm, assuming Non-Gaussian noise. Visualize the estimated DAG.

b) Based on the DAG estimated in (a), which variables are direct causes of travel time?

c) What is the total causal effect of cars on travel time according to estimated LiNGAM model?

d) How does the estimated linear SCM with non Gaussian noise look like?

e) Assume Marc traveled today from Zurich to Rome. He had to wait 40 minutes for the long-distance bus and his total travel time was 742 minutes. The actual driving time was 640 minutes and the traffic volume was 133. What would his travel time have been if the traffic volume would had only been 110?